



Submitted By : Maryam Rehan

Cyber Security Internship Report

Week 02

Module 1: Footprinting & Reconnaissance Attacks with Multiple Kali Tools

Module 2: Footprinting & Reconnaissance Attacks with GHDB

Module 3: Footprinting with Maltego

Module 4: Footprinting & Reconnaissance with theHarvester

Module 5: Network Scanning with Zenmap

Objective:

The objective of this lab is to understand and practically perform different cybersecurity footprinting, reconnaissance, information-gathering, and network-scanning techniques using tools such as Kali Linux reconnaissance tools, Google Hacking Database (GHDB), Maltego, theHarvester, and Zenmap. The lab focuses on collecting publicly available information, identifying domains, DNS records, web technologies, exposed information, and hosts within an authorized lab environment while documenting the results through commands, outputs, and screenshots.

Module 1: Footprinting & Reconnaissance Attacks with Multiple Kali Tools

➤ **Task 01 To 06**

Screen Shoots:

```
root@kali: ~/Module1
Session Actions Edit View Help
(root@kali)-[~]
# whoami
root
(root@kali)-[~]
# pwd
/root
(root@kali)-[~]
# mkdir -p ~/Module1
cd ~/Module1
(root@kali)-[~/Module1]
# pwd
/root/Module1
(root@kali)-[~/Module1]
#
```

```
root@kali: ~/Module1
Session Actions Edit View Help
(root@kali)-[~/Module1]
# whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeletePr
ohibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProh
ibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransf
erProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdatePr
ohibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wi
cf/
>>> Last update of whois database: 2026-09-26T01:46:50Z <<<
```

```
root@kali: ~/Module1
Session Actions Edit View Help
For more information on Whois status codes, please visit https://icann.org/epp
NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.
TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
```

```
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability. VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign
reserves the right to modify these terms at any time.
```

```
The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: networkwalks.com
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: https://www.godaddy.com
Updated Date: 2025-11-12T05:08:41Z
Creation Date: 2019-11-06T17:51:46Z
Registrar Registration Expiration Date: 2027-11-06T17:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1.4806242505
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Registry Registrant ID: Not Available From Registry
Registrant Name: Registration Private
Registrant Organization: Domains By Proxy, LLC
```

```
root@kali: ~/Module1
Session Actions Edit View Help
information about domain name registration records. Any use of this data for
any other purpose
is expressly forbidden without the prior written permission of this registrar
. By submitting
an inquiry, you agree to these terms and limitations of warranty. In particul
ar, you agree not
to use this data to allow, enable, or otherwise support the dissemination or
collection of this
data, in part or in its entirety, for any purpose, such as transmission by e-
mail, telephone,
postal mail, facsimile or other means of mass unsolicited, commercial adverti
sing or solicitations
of any kind, including spam. You further agree not to use this data to enable
high volume, automated
or robotic electronic processes designed to collect or compile this data for
any purpose, including
mining this data for your own personal or commercial purposes. Failure to com
ply with these terms
may result in termination of access to the Whois database. These terms may be
subject to modification
at any time without notice.

**NOTICE** This WHOIS server is being retired. Please use our RDAP service in
stead.

(root@kali)-[~/Module1]
#
```

Figures : WHOIS.png

```
(root@kali)-[~/Module1]
# whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpm_client
], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpm_client], I
P[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHead
ers[permissions-policy,x-redirect-by,upgrade]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.2], Cookies[__wpm_cl
ient], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Googl
e-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpm_client], IP[192.232
.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.2,WordPress Download Ma
nager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=
50511&pvtVar2=7&scvVar2=12,application/json,application/ld+json,module
e,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHea
ders[permissions-policy,link,upgrade], WordPress[7.1.2]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.2], Country[UNITED S
TATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, H
TTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPres
s 7.1.2,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Scr
ipt[4684NR-IPIB&pidnVar2=50511&pvtVar2=7&scvVar2=12,application/j
son,application/ld+json,module,speculationrules,text/javascript], Title[Netwo
rkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade], WordPress
[7.1.2]
```

Figure : WhatWeb.png

```
(root@kali)-[~/Module1]
# nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

Figure : NSLookup.png

```

(root@kali)-[~/Module1]
# curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=810a71846faa23529277498fe555c555; path=/; domain=networkwalks.com; secure; HttpOnly
content-type: text/html; charset=UTF-8
date: Sat, 26 Sep 2026 02:02:41 GMT
server: Apache

```

Figure : CURL.png

```

(root@kali)-[~/Module1]
# wafw00f https://networkwalks.com

```



~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit

```

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2

```

Figure : WAFW00F.png

```

(root@kali)-[~/Module1]
# dnsrecon -d networkwalks.com
2026-09-25T22:06:16.802342-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-25T22:06:16.803091-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-09-25T22:06:17.480738-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-09-25T22:06:18.018434-0400 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-09-25T22:06:18.817612-0400 INFO NS ns6136.hostgator.com 192.232.216.131
2026-09-25T22:06:19.336935-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-09-25T22:06:19.337209-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-09-25T22:06:19.862138-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-09-25T22:06:20.898903-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-09-25T22:06:21.206841-0400 INFO A networkwalks.com 192.232.216.135
2026-09-25T22:06:22.224961-0400 INFO TXT networkwalks.com v=spf1 +a +mx +

```

```
Session Actions Edit View Help
2026-09-25T22:06:20.898903-0400 INFO MX mail.networkwalks.com 192.232.216
.135
2026-09-25T22:06:21.206841-0400 INFO A networkwalks.com 192.232.216.135
2026-09-25T22:06:22.224961-0400 INFO TXT networkwalks.com v=spf1 +a +mx +
ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-09-25T22:06:22.225372-0400 INFO TXT networkwalks.com google-site-ver
ification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-09-25T22:06:22.879113-0400 INFO Enumerating SRV Records
2026-09-25T22:06:26.508171-0400 INFO SRV _autodiscover._tcp.networkwalks.
com cpanelmaildiscovery.cpanel.net 184.94.203.14 443
2026-09-25T22:06:26.508406-0400 INFO SRV _autodiscover._tcp.networkwalks.
com cpanelmaildiscovery.cpanel.net 184.94.203.11 443
2026-09-25T22:06:26.508541-0400 INFO SRV _autodiscover._tcp.networkwalks.
com cpanelmaildiscovery.cpanel.net 184.94.204.9 443
2026-09-25T22:06:26.508646-0400 INFO SRV _autodiscover._tcp.networkwalks.
com cpanelmaildiscovery.cpanel.net 184.94.203.9 443
2026-09-25T22:06:26.508758-0400 INFO SRV _autodiscover._tcp.networkwalks.
com cpanelmaildiscovery.cpanel.net 184.94.204.11 443
2026-09-25T22:06:26.508835-0400 INFO SRV _autodiscover._tcp.networkwalks.
com cpanelmaildiscovery.cpanel.net 184.94.204.14 443
2026-09-25T22:06:26.508900-0400 INFO SRV _autodiscover._tcp.networkwalks.
com cpanelmaildiscovery.cpanel.net 184.94.204.15 443
2026-09-25T22:06:26.508965-0400 INFO SRV _autodiscover._tcp.networkwalks.
com cpanelmaildiscovery.cpanel.net 184.94.203.15 443
2026-09-25T22:06:26.509978-0400 INFO 8 Records Found
2026-09-25T22:06:26.510292-0400 INFO Completed enumeration for domain: networ
kwalks.com
```

Figure : DNSRecon.png

Findings:

The reconnaissance activities provided publicly available information about networkwalks.com. WHOIS identified the domain registrar and registration dates. WhatWeb identified Apache, WordPress, WordPress Download Manager, jQuery and other web technologies. NSLookup resolved the domain to 192.232.216.135 using Google DNS. CURL confirmed an HTTP/2 200 response and revealed standard HTTP response headers. WAFW00F identified ModSecurity (SpiderLabs) as the detected web application firewall. DNSRecon identified the domain's SOA, name servers, MX, A and TXT/SPF records.

Module 2: Footprinting & Reconnaissance Attacks with GHDB

- **Task 01: Use GHDB to identify 10 publicly indexed security-camera-related entries and document their corresponding Google dorks**

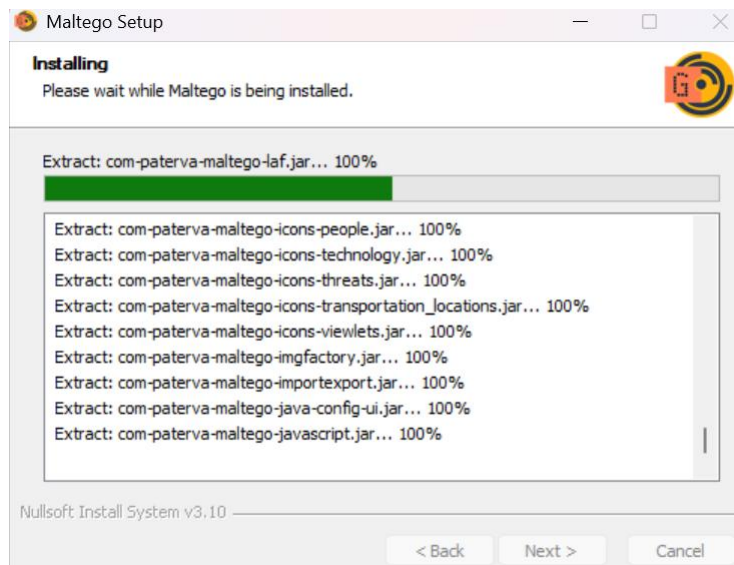
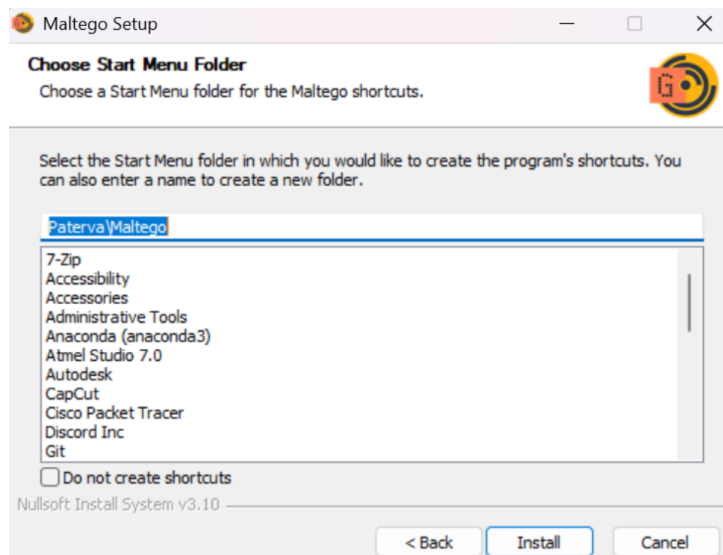
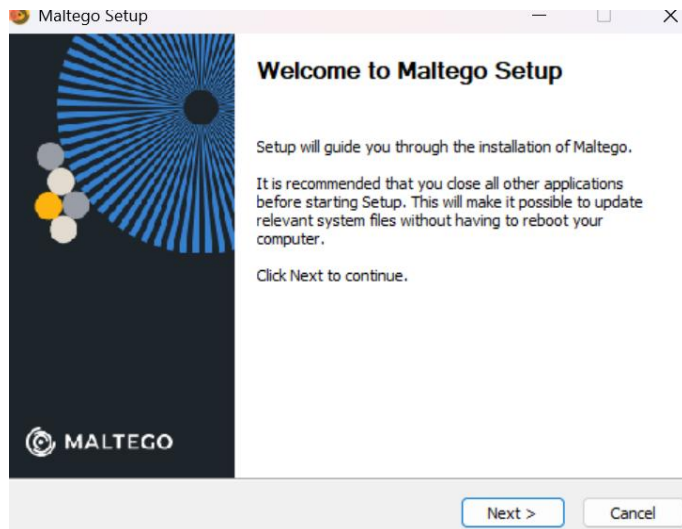
Sr. No.	GHDB Entry / Title	Google Dork / Search Query	Purpose / Description
1	Webcam Login	intitle:"webcam" "login"	Identifies publicly indexed webcam login pages.
2	EvoCam	intitle:"EvoCam" inurl:"webcam.html"	Identifies pages associated with EvoCam webcam software.
3	Webcam 7	intitle:"webcam 7" inurl:"/gallery.html"	Searches for indexed Webcam 7 gallery pages.
4	Axis Network Cameras	indexFrame.shtml	Identifies indexed Axis network-camera interface pages.
5	webcamXP 5	intitle:"webcamXP 5" inurl:8080 "Live"	Searches for indexed webcamXP 5 interfaces.
6	Brains, Corp. Camera	allintitle:Brains, Corp. camera	Searches for pages associated with Brains Corp. camera systems.
7	DVR Web Client	intitle:"DVR Web client"	Identifies indexed DVR web-client pages.
8	Video Web Server	intext:"Video Web Server"	Searches for pages containing the phrase "Video Web Server".
9	Yawcam	intitle:"yawcam" "It's a webcam!"	Identifies pages associated with Yawcam webcam software.
10	Active Webcam	intitle:"active webcam page"	Searches for pages associated with Active Webcam interfaces.

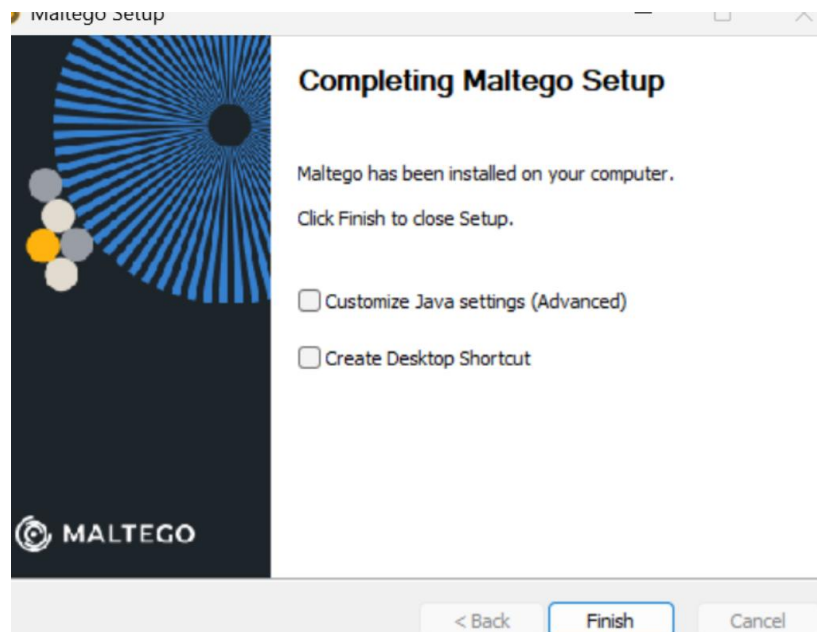
- **Task 02 : Used Google Dorking techniques to identify publicly available mathematics-related PDF resources from educational institutions and documented their titles, institutions, search query, and source domains**

Sr. No.	Mathematics PDF / Resource	Institution / Source	Google Dork Used	Source Domain
1	Everything You Always Wanted To Know About Mathematics	Carnegie Mellon University	site:edu filetype:pdf mathematics	math.cmu.edu
2	The History of Mathematics: An Introduction	University of Illinois Urbana-Champaign	site:edu filetype:pdf mathematics	illinois.edu
3	Street-Fighting Mathematics	Massachusetts Institute of Technology (MIT)	site:edu filetype:pdf mathematics	mit.edu
4	On Proof and Progress in Mathematics	University of Toronto	site:edu filetype:pdf mathematics	math.toronto.edu
5	The Princeton Companion to Mathematics	Rutgers University	site:edu filetype:pdf mathematics	math.rutgers.edu
6	Basic Set Theory	University of Houston	site:edu filetype:pdf mathematics	math.uh.edu
7	Exploring Practical Applications and Real-World Impact	UNM Digital Repository	site:edu filetype:pdf mathematics	digitalrepository.unm.edu
8	MATHEMATICS?	Western Carolina University	site:edu filetype:pdf mathematics	wcu.edu
9	An Introduction to Higher Mathematics	Whitman College	site:edu filetype:pdf mathematics	whitman.edu
10	Mathematics	Bemidji State University	site:edu filetype:pdf mathematics	bemidjistate.edu

Module 3: Footprinting with Maltego

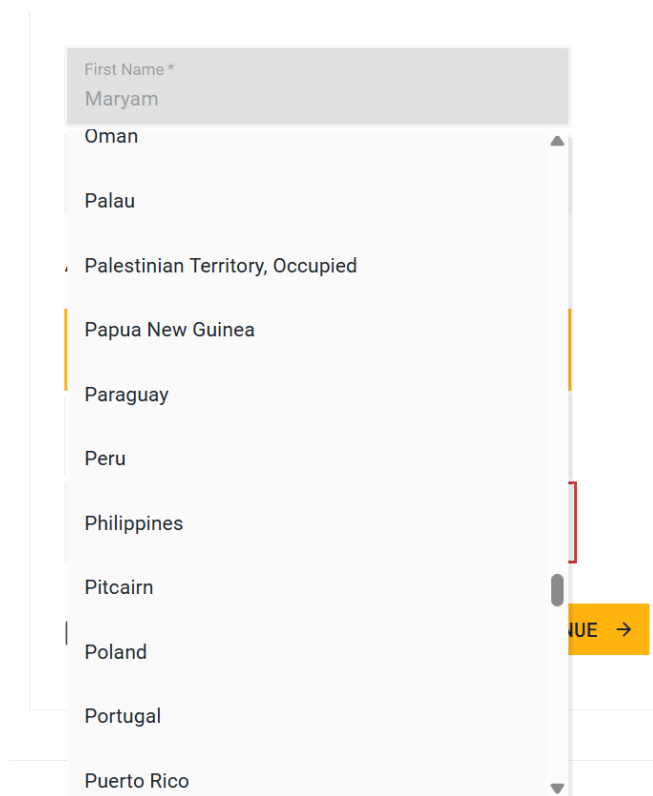
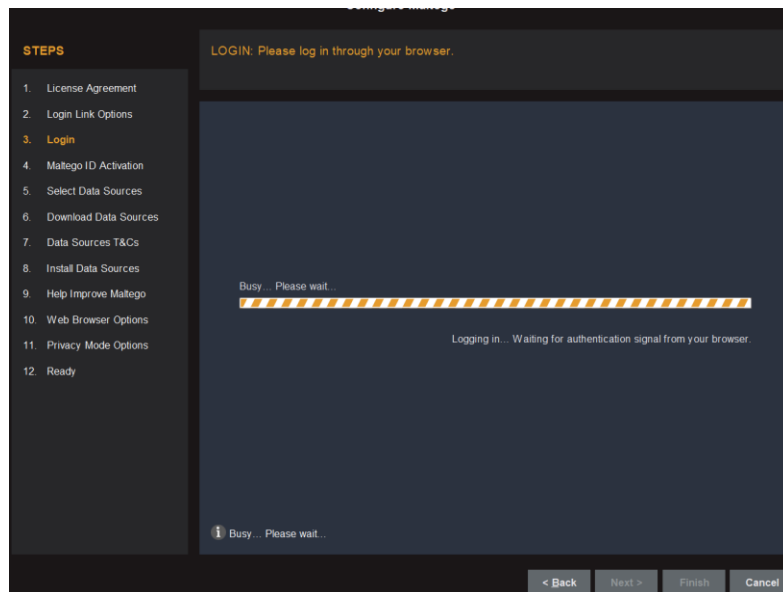
➤ Task 01: Windows PC Maltego Install & Account Setup





➤ Task 02: Email Address Harvesting from Target Domain





Issue Occur:

Maltego Graph was successfully downloaded and installed; however, during the Maltego ID registration process, Pakistan was not available in the country selection list and the registration form required country and phone-number information, so activation could not be completed without providing incorrect details. Therefore,

publicly accessible OSINT information related to the authorized target domain networkwalks.com was reviewed, and the email address info@networkwalks.com was identified on the Networkwalks website. No private or unauthorized information was accessed.

Sr. No.	Email Address	Target Domain	Source
1	info@networkwalks.com	networkwalks.com	Publicly accessible Networkwalks website

Observation:

The email address info@networkwalks.com was identified as a publicly available organizational contact email associated with the target domain networkwalks.com. The email address was visible on the publicly accessible Networkwalks website.

Evidence:

The screenshot shows info@networkwalks.com on the Networkwalks website, providing evidence that the email address is publicly associated with the target organization.

Note:

Only publicly available information was collected, and no private or unauthorized information was accessed.

Module 4: Footprinting & Reconnaissance with theHarvester

- **Task 01: Email and Subdomain Enumeration of Microsoft.com Using theHarvester**

```
(root@kali)~[Module1]
# theHarvester -d microsoft.com -b baidu -l 1000
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* [theHarvester]
*
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: microsoft.com

[*] Searching Baidu.

[*] No IPs found.

[*] No emails found.
```

Figure 01: TheHarvester reconnaissance results for microsoft.com using Baidu

TheHarvester was used to perform authorized reconnaissance on the target domain microsoft.com using Baidu as the search source with a limit of 1000 results. The search completed successfully; however, no IP addresses, email addresses, people, or hostnames were returned by the selected source. The result demonstrates that theHarvester can be used to collect publicly available reconnaissance information, although the availability of results depends on the selected search source.

➤ Task 02: Reconnaissance Using All Available Sources

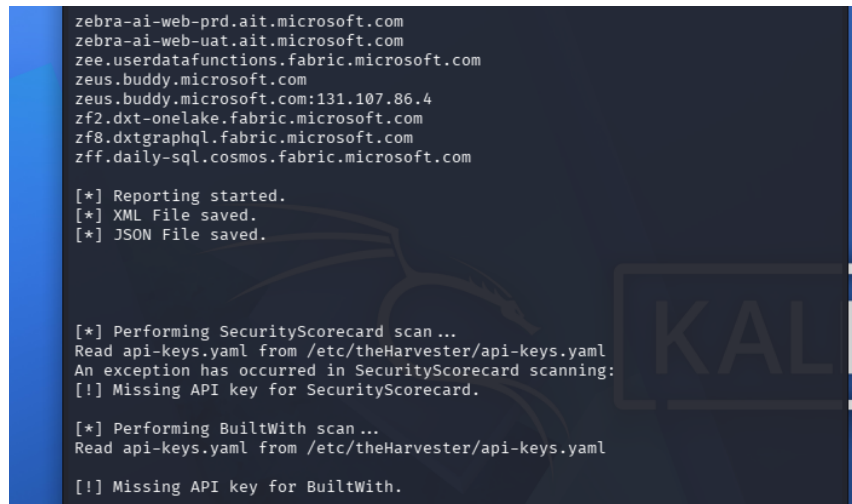
```
Session Actions Edit View Help
za0.daily-c.fabric.microsoft.com
zai-prod-mwh-proc.ait.microsoft.com
zai-prod-sec-proc.ait.microsoft.com
zai-uat-mwh-proc.ait.microsoft.com
zai-uat-sec-proc.ait.microsoft.com
zb2.dxt-c.fabric.microsoft.com
zb2.msituserdatafunctions.fabric.microsoft.com
zc6.msit-dfs.fabric.microsoft.com
zebra-ai-api-prd.ait.microsoft.com
zebra-ai-api-uat.ait.microsoft.com
zebra-ai-web-prd.ait.microsoft.com
zebra-ai-web-uat.ait.microsoft.com
zeus.buddy.microsoft.com
zeus.buddy.microsoft.com:131.107.86.4
zf0.userdatafunctions.fabric.microsoft.com

[*] Performing SecurityScorecard scan...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
An exception has occurred in SecurityScorecard scanning:
[!] Missing API key for SecurityScorecard.

[*] Performing BuiltWith scan...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for BuiltWith.
```

Figure 02: TheHarvester reconnaissance results for microsoft.com using all available sources

TheHarvester was executed against the authorized target domain microsoft.com using all available sources with a limit of 50 results. The scan successfully returned multiple publicly available Microsoft-related hosts and subdomains. Some integrated sources, including SecurityScorecard and BuiltWith, could not be queried because their required API keys were not configured. The scan nevertheless completed successfully and produced reconnaissance results from the available sources.



```
zebra-ai-web-prd.ait.microsoft.com
zebra-ai-web-uat.ait.microsoft.com
zee.userdatafunctions.fabric.microsoft.com
zeus.buddy.microsoft.com
zeus.buddy.microsoft.com:131.107.86.4
zf2.dxt-onelake.fabric.microsoft.com
zf8.dxtgraphql.fabric.microsoft.com
zff.daily-sql.cosmos.fabric.microsoft.com

[*] Reporting started.
[*] XML File saved.
[*] JSON File saved.

[*] Performing SecurityScorecard scan...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
An exception has occurred in SecurityScorecard scanning:
[!] Missing API key for SecurityScorecard.

[*] Performing BuiltWith scan...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for BuiltWith.
```

Figure 03: TheHarvester results saved in XML and JSON formats

TheHarvester was executed against microsoft.com using all available sources with a limit of 50 results. The scan returned multiple Microsoft-related hosts and subdomains. The results were successfully saved in XML and JSON formats. Some sources, including SecurityScorecard and BuiltWith, could not be queried because their API keys were not configured; however, the overall reconnaissance scan completed successfully.

NOTE:

Some third-party data sources required API keys that were not configured; therefore, those specific sources could not be queried. Other available sources returned reconnaissance results successfully.

Module 5: Network Scanning with Zenmap

➤ Task 1: Download & install Zenmap on Windows

Microsoft Windows binaries

Please read the [Windows section](#) of the Install Guide for limitations and installation instructions for the Windows version of Nmap. It's provided as an executable self-installer which includes Nmap's dependencies and the Zenmap GUI. We support Nmap on Windows 7 and newer, as well as Windows Server 2008 R2 and newer. We also maintain a [guide for users who must run Nmap on earlier Windows releases](#).

Note: The version of Npcap included in our installers may not always be the latest version. If you experience problems or just want the latest and greatest version, download and install [the latest Npcap release](#).

Latest stable release self-installer: [nmap-7.991-setup.exe](#)
Latest Npcap release self-installer: [npcap-1.88.exe](#)

We have written [post-install usage instructions](#). Please [notify us](#) if you encounter any problems or have suggestions for the installer.

Check the components you want to install and uncheck the components you don't want to install. Click Next to continue.

Select components to install:

Component	Description
☒ Nmap Core Files	Position your mouse over a component to see its description.
☒ Register Nmap Path	
☒ Npcap 1.00	
☒ Network Performance Imp	
☒ Zenmap (GUI Frontend)	
☒ Ndiff (Scan comparison to	
☒ Ncat (Modern Netcat rein	
☒ Nping (Packet generator)	

Space required: 84.2MB

Microsoft Install System v2.51

< Back Next > Cancel

nmmap Setup

Installing

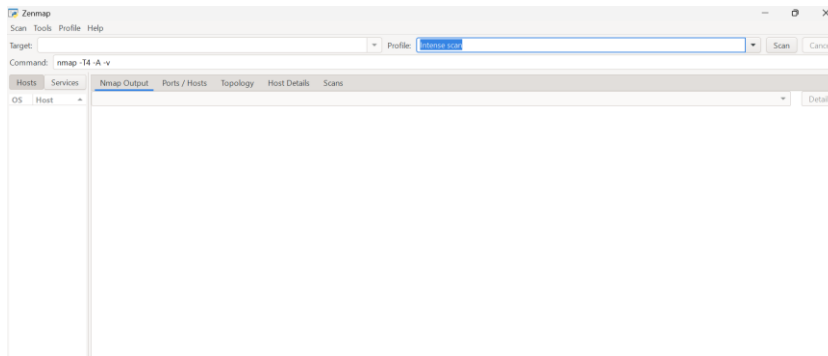
Please wait while Nmap is being installed.

Execute: "C:\Users\MT\AppData\Local\Temp\nsjE4DD.tmp\npcap-1.88.exe" /loopback_support

Show details

Microsoft Install System v3.11

< Back Next > Cancel



➤ Task 02: Find Local IP Address & LAN Subnet

- **IPv4 Address:** 192.168.0.109
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.0.1
- **LAN Subnet:** 192.168.0.0/24

```
Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::f2b4:8f55:21dc:8092%19
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::759a:633:a45f:3a93%13
    IPv4 Address. . . . . : 192.168.230.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
```

```
Ethernet adapter VMware Network Adapter VMnet8:

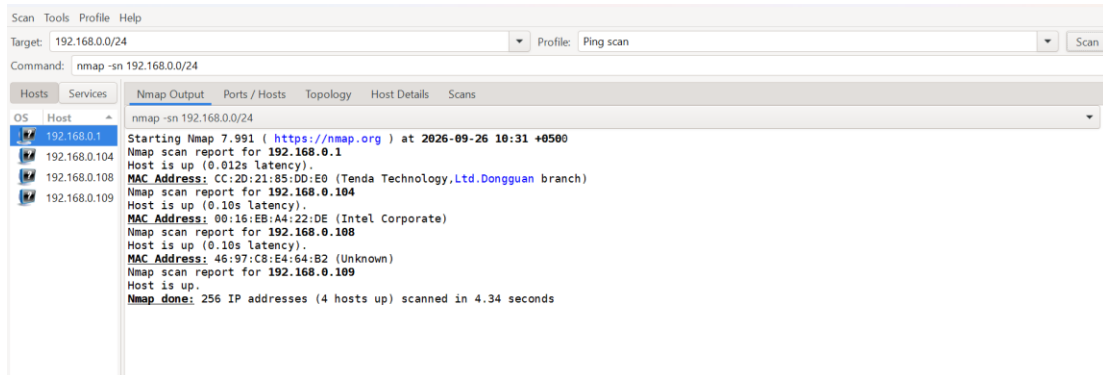
    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::858a:e47e:94ac:507b%11
    IPv4 Address. . . . . : 192.168.244.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : www.mtlinkwifi.com
    Link-local IPv6 Address . . . . . : fe80::1eef:3349:a638:cf99%14
    IPv4 Address. . . . . : 192.168.0.109
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.0.1

C:\Users\MT>
```

➤ Task 03: Find List of Live Hosts/PCs in IP Subnet



Zenmap was used to perform a Ping Scan on the authorized local subnet 192.168.0.0/24 using the command `nmap -sn 192.168.0.0/24`. The scan identified four live hosts on the local network.

1	192.168.0.1
2	192.168.0.104
3	192.168.0.108
4	192.168.0.109

➤ Task 04: How Many Hosts Are Live in the Subnet?

The Zenmap Ping Scan identified 4 live hosts in the local subnet 192.168.0.0/24. Nmap scanned 256 IP addresses and detected 4 hosts that were active and responding during the scan.

Result: 4 live hosts were identified.

➤ Task 05: IP Addresses of Live Hosts

The Zenmap Ping Scan identified the following active hosts within the authorized local subnet 192.168.0.0/24

1	192.168.0.1
2	192.168.0.104
3	192.168.0.108
4	192.168.0.109

These addresses represent the hosts that responded to the Ping Scan during the reconnaissance activity.

```
MAC Address: CC:2D:21:85:DD:E0 (Tenda Technology,Ltd.Dongguan branch)
Nmap scan report for 192.168.0.104
Host is up (0.10s latency).
MAC Address: 00:16:EB:A4:22:DE (Intel Corporate)
Nmap scan report for 192.168.0.108
Host is up (0.10s latency).
MAC Address: 46:97:C8:E4:64:B2 (Unknown)
Nmap scan report for 192.168.0.109
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 4.34 seconds
```

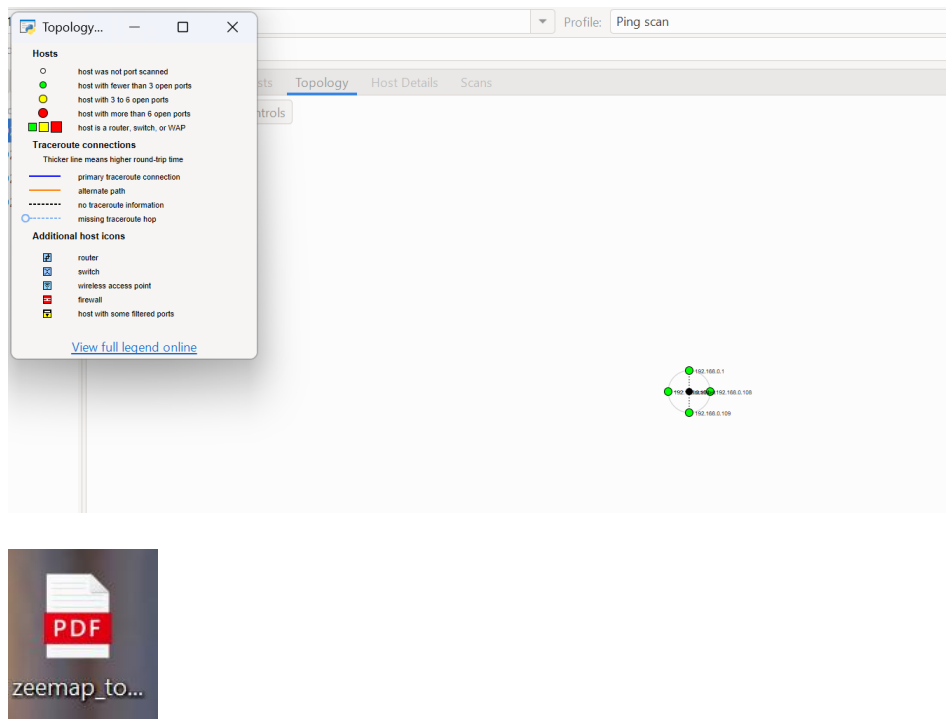
➤ Task 06: MAC Addresses of Live Hosts

During the Zenmap Ping Scan, MAC addresses were identified for three of the four detected live hosts. The MAC address of the local system (192.168.0.109) was not displayed in the Nmap output.

IP Address	MAC Address	Vendor
192.168.0.1	CC:2D:21:85:DD	Tenda Technology
192.168.0.104	00:16:EB:A4:22	Intel Corporate
192.168.0.108	46:97:C8:E4:64	Unknown
192.168.0.109	Not displayed	—

The results demonstrate that Zenmap/Nmap can identify MAC addresses and associated vendor information for hosts where the required local network information is available.

➤ Task 07: Display & Save Output Topology as PDF



Conclusion:

This Week 02 Cybersecurity & Ethical Hacking project provided practical experience in footprinting, reconnaissance, OSINT, information gathering, and network scanning. Different tools and techniques were used, including Kali Linux reconnaissance tools, Google Hacking Database (GHDB), Maltego, theHarvester, and Zenmap. Publicly available information such as domain details, DNS records, web technologies, indexed resources, email information, and subdomains was examined in an authorized and controlled environment. Zenmap was also used to identify live hosts, IP addresses, MAC addresses, and visualize the local network topology. Overall, the project improved practical understanding of reconnaissance methodologies and demonstrated how cybersecurity tools can be used responsibly for information gathering and network analysis.

